

Additional Results of "Towards Sparse Spiking LLMs: A Novel ANN-to-SNN Conversion Framework with Temporal Masking"

Table 1: (Result for **Reviewer d3yL,7NnQ**: Results on more models) Results on LLaMA-2-13B. "cal" denotes the SNN performance after calibration. "Avg. Acc." indicates the average zero-shot accuracy on four reasoning tasks. The percentage (X%) refers to the proportion of activations covered by the temporal masks.

Method	Precision	WinoGrande	ArcC	ArcE	PIQA	Avg.Acc	Spike activation ratio (↓)
Baseline	FP16	72.22	49.06	77.48	80.52	70.20	-
MTH (0%)	T=4, N=10	72.14	50.85	77.57	80.36	70.23	69.93%
Ours (40%)	T=4, N=10	69.06	42.58	68.94	75.52	64.02	51.71%
Ours.cal (40%)	T=4, N=10	71.19	43.34	71.34	77.58	65.86	48.79%
Ours (50%)	T=4, N=10	60.85	30.46	51.56	68.06	52.73	43.82%
Ours.cal (50%)	T=4, N=10	67.64	40.10	68.94	77.42	63.53	39.70%

Table 2: (Result for **Reviewer d3yL,7NnQ**: Results on more models) Results on Qwen-7B. "cal" denotes the SNN performance after calibration. "Avg. Acc." indicates the average zero-shot accuracy on four reasoning tasks. The percentage (X%) refers to the proportion of activations covered by the temporal masks.

Method	Precision	WinoGrande	ArcC	ArcE	PIQA	Avg.Acc	Spike activation ratio (↓)
Baseline	FP16	72.38	49.91	74.71	81.23	69.55	-
MTH (0%)	T=4,n=10	71.90	51.88	77.44	79.76	70.24	84.21%
Ours (40%)	T=4, N=10	67.56	50.51	76.68	78.35	68.27	57.96%
Ours.cal (40%)	T=4, N=10	69.06	50.85	76.89	78.07	68.71	56.52%
Ours (50%)	T=4, N=10	59.27	40.19	63.80	72.58	58.96	49.46%
Ours.cal (50%)	T=4, N=10	62.75	43.00	66.46	74.70	61.72	47.21%

Table 3: (Result for **Reviewer d3yL**: Ablation studies on probabilistic masking vs. deterministic masking) Ablation results of our method using probabilistic/deterministic mask on LLaMa-2-7B with $T = 4$ and $n = 10$

Method	WinoGrande	ArcC	ArcE	PIQA	Avg.Acc	Ratio (↓)
Ours (40%)	68.35/66.93	41.64/42.75	67.76/67.51	75.90/75.90	63.41/63.27	45.95%/51.67%
Ours.cal (40%)	68.11 / 67.25	46.93 /43.69	73.61 /69.82	78.40 /76.61	66.76 /64.34	44.83%/50.77%
Ours (50%)	61.88/59.75	36.26 /36.60	58.21/57.45	72.85/72.52	57.30/56.58	38.13%/44.22%
Ours.cal (50%)	67.09 /64.88	44.28 /37.46	71.63 /62.71	77.91 /74.21	65.23 /59.80	36.58%/42.38%